

Aejandro De La Torre

LIS 440

Lecture: Week 14 | Pandas Aggregation, Data Cleaning, and Intro to Visualization

12/2/25

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output'
code = root / 'code'

# -----
# Data Cleaning
# -----

# Fetch the data
icecreamDF = pd.read_csv(data_processed / 'icecreamNA.csv')
print() # return
print("Ice Cream Data: ")
print(icecreamDF)

# Use this method to drop NA values
# print()
# print("Drop NA Data:")
# icecreamCleanDF = icecreamDF.dropna() # to drop N/A values

# Fill the NA values with 0
print()
print("Zero Out the NA data:")
icecreamCleanDF = icecreamDF.fillna(0)
```

```

print(icecreamCleanDF)
# If want to overwrite the original df directly just
# icecreamDF.fillna(0, inplace=True)

# Method to check for NA data
print()
print("Look for NA data:")
icecreamNAvaluesDF = icecreamDF.isna()
print(icecreamNAvaluesDF)

# We can inspect the total NA values...
print()
print("Total NA values:")
totalNAvalues = icecreamNAvaluesDF.sum()
print(totalNAvalues)

# Alternatively we can combine the check and inspect steps
print()
print("Total NA values (method chaining)")
totalNAvalues = icecreamDF.isna().sum()
print(totalNAvalues)

```

Shell

```

>>> %Run icecream_datatype_analysis.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 14/Tuesday/code

```

Ice Cream Data:

	Flavor	SugarCone	Cup	WaffleCone
0	Chocolate	44	34	26
1	Strawberry	22	41	22
2	Vanilla	33	31	44
3	Blue Moon	11	29	22
4	Orange	32	32	11
5	Mnookidough	23	22	52
6	Cookie Dough	41	11	20

Zero Out the NA data:

	Flavor	SugarCone	Cup	WaffleCone
0	Chocolate	44	34	26
1	Strawberry	22	41	22
2	Vanilla	33	31	44
3	Blue Moon	11	29	22
4	Orange	32	32	11
5	Mnookidough	23	22	52
6	Cookie Dough	41	11	20

Look for NA data:

	Flavor	SugarCone	Cup	WaffleCone
0	False	False	False	False

1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False
5	False	False	False	False
6	False	False	False	False

Total NA values:

Flavor 0

SugarCone 0

Cup 0

WaffleCone 0

dtype: int64

Total NA values (method chaining)

Flavor 0

SugarCone 0

Cup 0

WaffleCone 0

dtype: int64

>>>